卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0306 30-B003701S-T0306

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 03 卷 第 06 册
Structure SPECIALITY VOLUME BOOK

卷册名称 直流接地极线构架加工图
Volume name working drawings of frame of grounding electrode line

批 准 APPROVED BY	张[illegible]
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	庞金来
图 纸 7 张 DRAWING NOS.	打印件 1 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0306-001	钢结构加工说明 specification for fabrication and design of steel works	A	1		
2	30-B003701S-T0306-002	直流接地极线构架透视图 perspective drawing of frame of grounding electrode line	A	1		
3	30-B003701S-T0306-003	直流接地极线构架腿身结构图(1/2) ① extension leg for frame of grounding electrode line (1/2) ①	A	1		
4	30-B003701S-T0310-004	直流接地极线构架腿身结构图(2/2) ① extension leg for frame of grounding electrode line (2/2) ①	A	1		
5	30-B003701S-T0306-005	直流接地极线构架腿身结构图 ② extension leg for frame of grounding electrode line ②	A	1		
6	30-B003701S-T0306-006	直流接地极构架身部结构图 ③ superstructure for frame of grounding electrode line ③	A	1		
7	30-B003701S-T0306-007	直流接地极构架梁结构图 ④ beam structure for frame of grounding electrode line ④	A	1		

注:

SPECIFICATION FOR FABRICATION AND DESIGN OF STEEL WORKS

1. GENERAL NOTES:

All dimemsions are in millimeters unless otherwise noted.

The elevations are in meters.

All steel members should adopt the anti-corrosive treatment of hot dip galvanizing,the galvanized standard shall conform to the latest revision of ASTM-A123M & ASTM-A153M.

2.MATERIALS:

Structural steel:

The grade of steel used in the structure shall be Q345B(Fy=345Mpa ,Fu=470Mpa) or Q235B(Fy=235Mpa ,Fu=375Mpa).

The steel shall confirm to《GB/T 1591-2008 High strength low alloy structural steels》

Structural bolts, nuts and washers:

Bolts shall be of steel grade 6.8 or grade 8.8, hot dip galvanized, with match nuts and washers;

Nuts and washers shall be of steel grade 6.8 or grade 8.8, hot dip galvanized.

The bolts and nuts shall confirm to《DL/T 284-2012 Hot-dip galvanized bolts and nuts for tower and hardware of transmission lines》

Welding rod:

The type of welding rod shall match with the steel.

For Q345B steel, the type is E50xx(F_{E50}=490Mpa);

For Q235B steel, the type is E43xx(F_{E43}=420Mpa);

3.MANUFACTURE:

Only after lofting, all members and gusset plates can be cut in the manufactory. The sizes of members listed in the steel schedule are only used for the preparation of steel.

During the fabrication process of steel gantry tower and girder members, all members shall be pre-assembled before producing in batches.

Only after steel members of gantry tower and girder must match the requirements of the relevant design drawings and standards, they can be transported to the construction-site.

Welding shall be finished before hot dip galvanizing.

In order to avoid bending, the hanging points shall be properly selected while galvanized steel members are taken out from the galvanization caldron,

All production welding shall be carried out in accordance with the requirements of welding.

The leg size of fillet welds shoud not be less than the thickness of the thinner part welded, but not more than 1.2 times that.

For the thick plate , grooving should be adopted in order to ensure the strength of connection.

The spacing between ends of angles shall be 10mm.

For the type of inside splice angle, the angle of splice angle will be filleted.

For the type of outer splice angle, the inside arc of splice angle will be cut into right-angle.

The standard dimensions of steel angle neat line and bolt, max.allowable diameter of bolt hole should be taken from Table 1, where not otherwise noted in the drawing .

The standard spacing between centres of bolts , edge and end distances on steel angles and gussets should be taken from Table 2, where not otherwise noted in the drawing.

the component shall adopt the technics to stop the lamellar tearing if the thickness of fillet weld is more than 20mm.

The bolt at the column feet adopts double nuts.

4.TRANSPORTATION AND STACKING:

During the transportation, the same kind of members should be tied up into bundles. To avoid the deformation of the members and the damage of galvanized layer due to mutual collision, the sleepers should be lined at the bottom of the members and between their layers.

After the transporting the membes to the construction-site, they should be piled separately according to the numbe of gantry tower and girder to avoid confusion during assembly.

The damaged part of zinc layer, which is caused by transportation and hoisting, should be cleaned and then brushed with zinc rich paint.

5.After erection of gantry,the holding-down pad shall be welded with the leg slab ,then brushed with zinc rich paint.

6.All steels should be rolled steel and shall confirm to local standards.

7.The fabricator should review all the material that is used in the fabrication of the entire structure, all mill test reports for material compliance, all material suppliers for their manufacturing procedures and quality control programs, and all welding electrodes.

8.The purchase should review and agree on the fabricator's materials specifications, supply sources, material identification, storage, traceability procedures, and certified mill test reports

9.welding

1).the fabricators' welding Procedures and welder-certification programs should be in accordance with the latest revision of the American welding society's Structural Welding Code for Steel (AWS D1 1/D1 1M).

2).the purchaser should establish requirements for the review of and agreement on the fabricators' welding procedures for various types of welds and their welding certification programs.

10.Structure Coating

1).where galvanizing is required, the procedure and facilities should be agreed on by the purchaser and the fabricator.

2).After hot-dip galvanizing, nondestructive testing may be specified to ensure that there have been no adverse changes to the finished product.

3).The magnetic thickness measurement method, is usually the test that is used to determine the thickness of the zinc coating.

1. 总说明:

除特殊注明外，所有标注尺寸单位为毫米.

标高标注的单位为米.

所有钢构件采用热浸锌防腐处理. 热镀锌标准应符合最新版规范ASTM-A123M以及ASTM-A153规定.

2. 材料:

钢材:

钢结构所用的钢材为Q345B(Fy=345Mpa ,Fu=470Mpa)或Q235B(Fy=235Mpa ,Fu=375Mpa).

钢材需要满足《低合金高强度结构钢(GB/T 1591-2008)》的要求.

螺栓，螺母和垫圈:

螺栓应为6.8或8.8级的热浸锌件，带有螺母、平垫圈;

螺母、垫圈应为相应的热浸锌件.

螺栓螺母需满足《 输电线路杆塔及电力金具用热浸镀锌螺栓与螺母(DL/T 284-2012) 》要求.

焊条:

焊条类型应与连接母材材质匹配.

对于Q345钢材，采用E50xx型焊条(F_{E50}=490Mpa).

对于Q235钢材，采用E43xx型焊条(F_{E43}=420Mpa).

3. 加工制造:

所有构件及节点板均需放样后下料，钢材明细表中的构件尺寸仅供备料用.

钢结构加工过程中必须进行试组装，然后才能成批加工.

钢结构加工必须满足相关图纸和验收规范要求方能运输到现场.

焊接构件应在完成焊接后再镀锌.

为避免镀锌构件从锌锅中取出时发生弯曲，应选择合适的吊点.

构件焊接应按照相应的焊接规程、规范的要求进行.

角焊缝高度不小于连接构件的最小厚度，但不大于1.2倍的连接构件的最小厚度.

对于较厚的焊件需采用坡口焊，保证连接强度.

两角钢对接接头处的间隙为10mm.

内包角钢连接时，采用铲内包角钢背的方法，外包角钢时采用铲外包角钢弧的方法.

角钢基准线和螺栓准线除图中注明外，应按表1采用.

角钢及节点板的螺栓间距、边距除图中注明外，应按表2采用.

柱脚锚栓采用双螺母.

4. 运输与堆放:

运输期间，同类构件应捆绑成束堆放在一起. 枕木放在构件末端和层间以防止变形和由于相互碰撞带来镀锌层的破坏.

构件运输到现场后，桁架构件应分离堆放，以防组装期间带来混淆.

由于运输和吊装破坏的镀锌层部分，应先清洗，然后涂刷富锌漆.

5. 构架安装完毕后，地脚螺栓垫板与柱脚焊接，刷富锌漆防腐.

6. 所有钢材均为轧制，且应符合当地标准.

7.制造商应审查所有应用于整个结构的制作的材料、所有出厂试验报告材料的符合性、所有材料供应商的制造程序和质量控制程序，以及所有焊条质量.

8.买方应审查制造商提供的材料规格、供应来源、材料的识别、存储、可追溯性程序、材料质检报告.

9.焊接

1).制造商的焊接工艺和焊工资格认证程序需满足现行美国焊接协会《钢结构焊接规范AWS D1 1M》的要求.

2).采购部应对制造商的各类焊缝焊接认证程序及要求进行审查.

10.结构涂层

1).热浸锌防腐工艺以及实施程序应该由买方和制造厂同意.

2).热浸锌完成后，需指定无损检测，以确保对成品没有不利的变化.

3).通常采用磁厚度测量法测定镀层厚度.

TABLE 1

No. 编号	FLANGE DIMENSION 角钢肢宽 b	NEAT LINE DIMENSION 基准规线距 a1	BOLT DIMENSION 螺栓准线距 SINGLE LINE 单排 a0	DOUBLE LINES 双排 a1	DOUBLE LINES 双排 a2	MAX. ALLOWABLE DIAMETER OF BOLT HOLE 最大允许开孔直径 Ø
1	50	25	25			17.5
2	56	28	28			
3	63	30	30			
4	70	35	35			21.5
5	75	38	38			
6	80	40	40			
7	90	45	45			
8	100	50	50			
9	110	55	55	45	75	
10	125	60	60	50	85	
11	140	70	70	55	90	25.5
12	160	80	80	60	105	
13	180	90	90	65	120	
14	200	100	100	75	135	

Note:
1.The unit of all values listed in the table is millimeter.
2.Single nut bolt includes one standard nut, one plain washer, one spring washer.
3.Double nut bolt includes one standard nut, one thin nut, one plain washer.

说明:
1.表格中所有数值均以毫米为单位.
2.单帽螺栓包含一标准帽，一平垫圈，一弹簧垫圈.
3.双帽螺栓包含一标准帽，一薄帽，一平垫圈.

TABLE 2

NOMINAL DIA. OF BOLT 螺栓直径	DIA. OF HOLE 螺栓孔直径	DISTANCE BETWEEN CENTRES 螺栓孔距 SINGLE LINE 单排	DISTANCE BETWEEN CENTRES 螺栓孔距 DOUBLE LINES 双排	EDGE AND END DISTANCE 螺栓边距及端距 Ld	Lz	La
16	17.5	50	80	25	≮21	≮23
20	21.5	60	100	30	≮26	≮28
24	25.5	80	120	40	≮31	≮33

Note: 1.the unit of all values listed in the table is millimeter.
说明: 1.表格中所有数值均以毫米为单位.
2."Lz" is the edge distance for a rolled edge.
2."Lz" 表示轧制边距.
3."La" is the edge distance for a sheared edge.
3."La" 表示切角边距.
4."Ld" is the end distance.
4."Ld"表示端距.

BILLS OF BOLT TYPE
螺栓类型明细表

TYPE 类型	THROUGH THICK 通过厚度
M16X40	8~12
M16X50	13~22
M16X60	23~32
M20X45	10~15
M20X55	16~25
M20X65	26~35
M20X75	36~45
M20X85	46~55
M20X95	56~65
M24X55	16~20
M24X65	21~30
M24X75	31~40

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

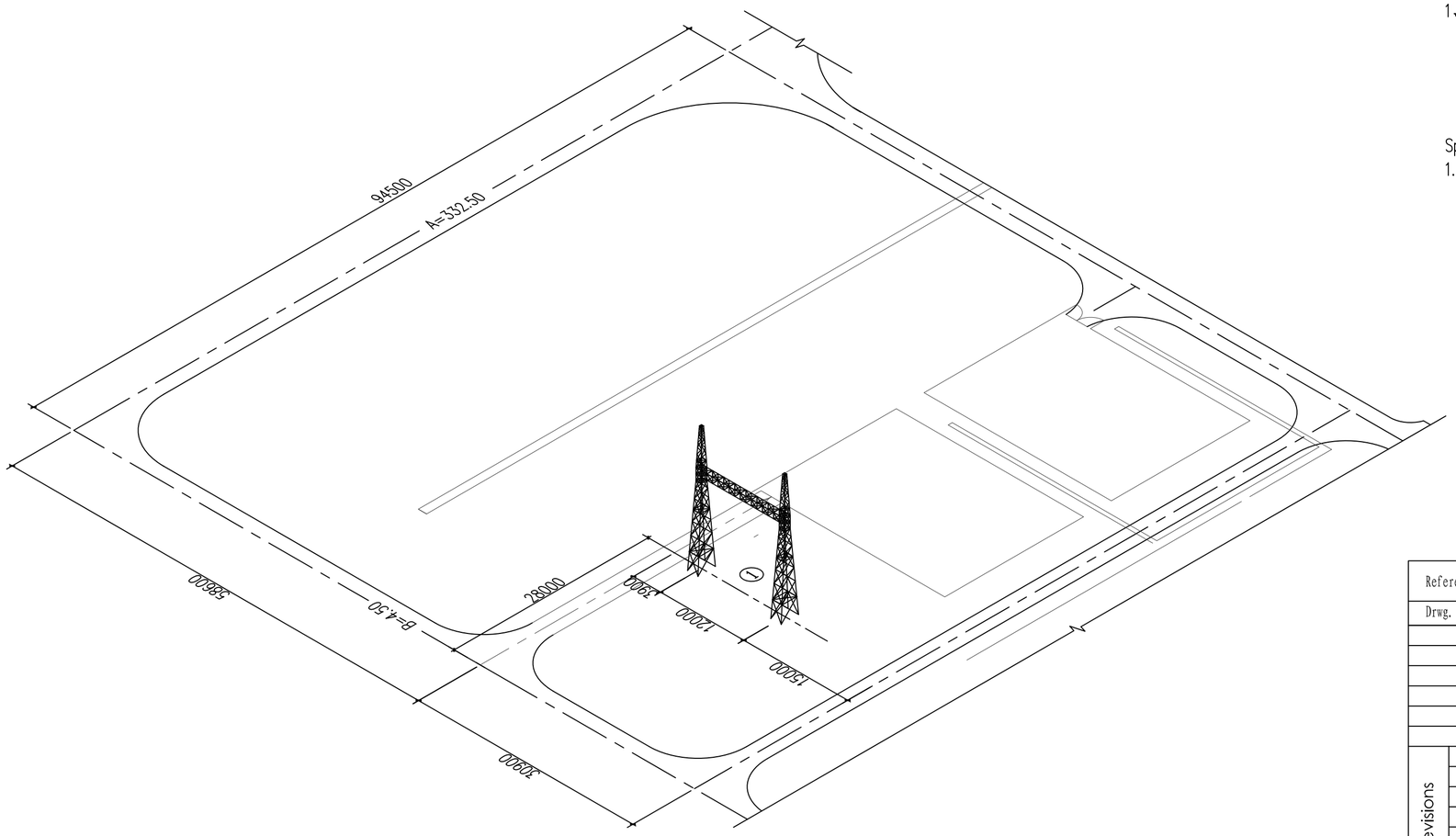
M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	施金来	18.11.20
校核 Checked	毛惠平 古韶斌	18.11.20
审核 Reviewed	郑伟清	18.11.20
批准 Approved	韩加亭	18.11.20
比例 Scale		图号 DWG NO. 30-B003701S-T0306-001

钢结构加工说明
SPECIFICATION FOR FABRICATION AND DESIGN OF STEEL WORKS

版本 REV. A

420X594



接地极线构架 layout of frame of grounding electrode line

说明：

1、场地±0.000相当于绝对标高193.40(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。

Specification :

1. The site level of ±0.000 is equal to the absolute elevation of193.40 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.

LISTS OF steelworks 钢结构工程量

NO. 编号	NAME 代号	QTY 数量	DRAWING NO. 施工图号	PCS单个/t	TOTAL总计/t
①	FRAME OF GROUNDING ELECTRODE LINE	1	30-B003701S-T0306-003~007	3.14	3.14

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	龙金来	18.11.20
校核 Checked	王惠千 古超城	18.11.20
审核 Reviewed	邹建清	18.11.20
批准 Approved	杨立亭	18.11.20
比例 Scale	1:400	图号 DWG NO.

直流接地极线构架透视图

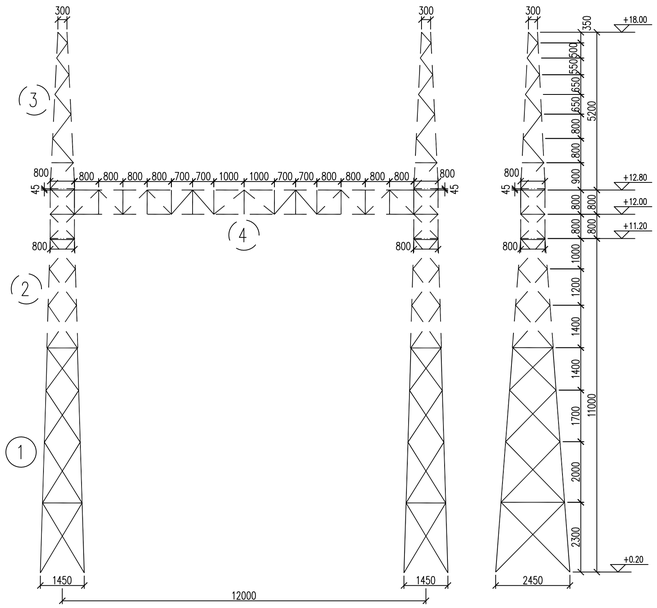
perspective drawing of frame of grounding electrode line

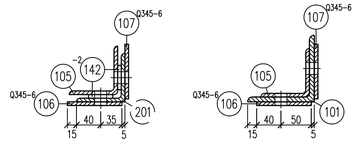
30-B003701S-T0306-002

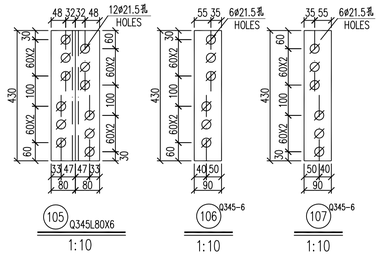
版本 REV. A

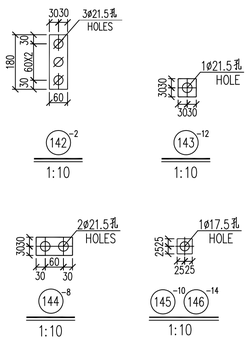
专业 Speciality	会签 Sign	日期 Date
变电一次	郑煜	18.11.20

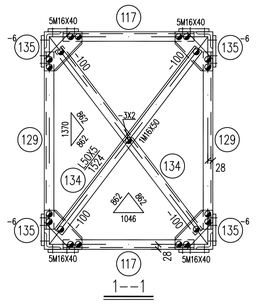
420X594

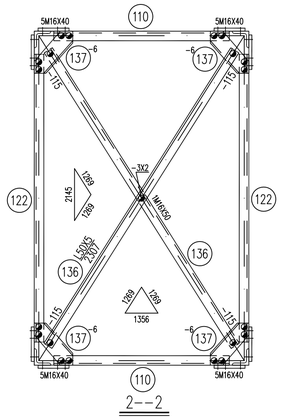


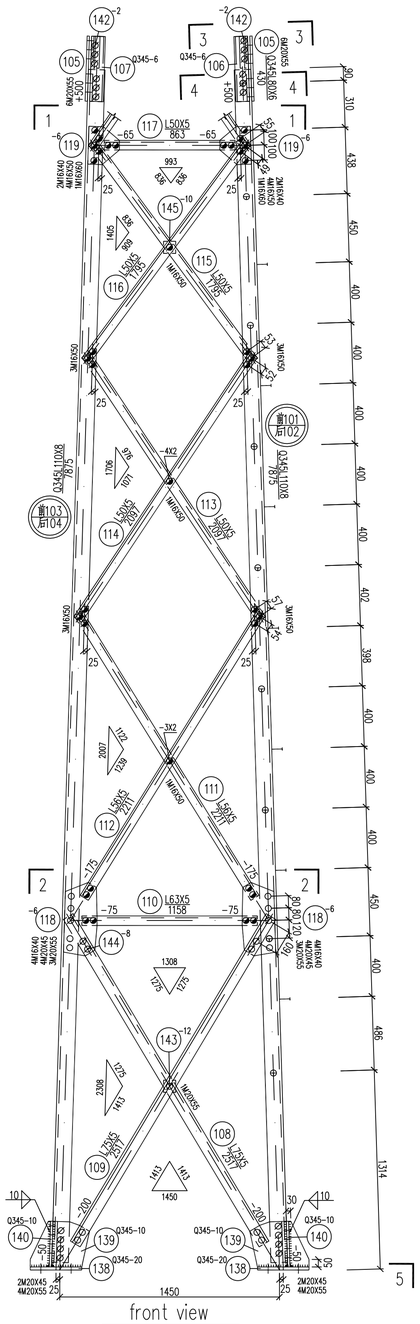


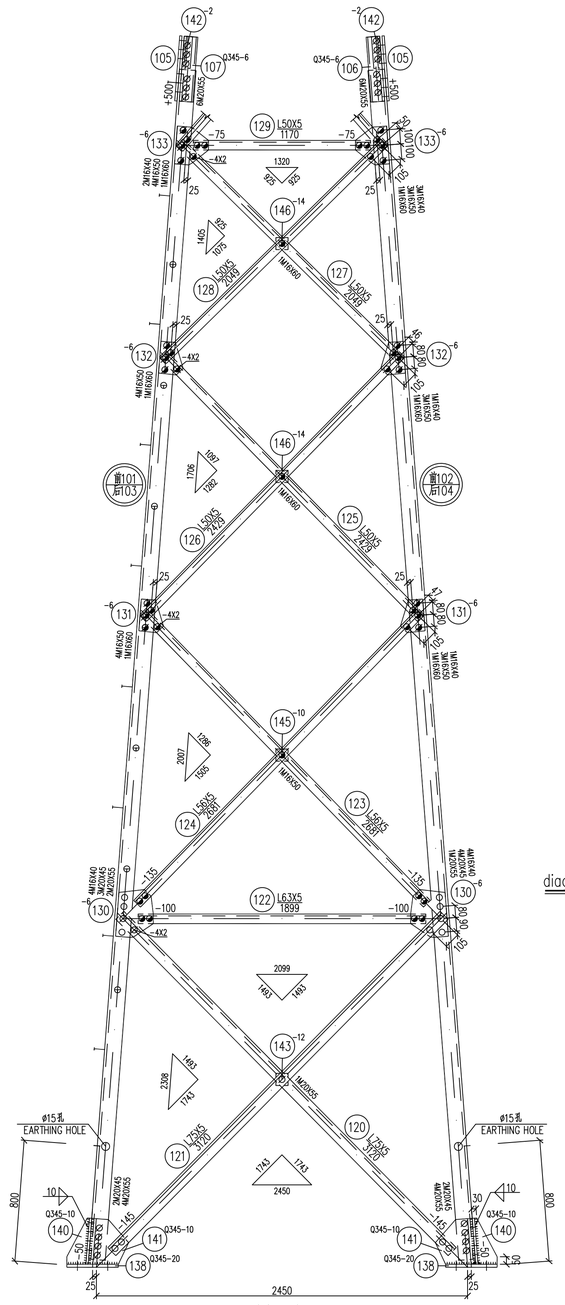












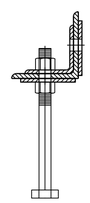


diagram of step-bolt in bolt hole

螺栓、脚钉、垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备注 REMARK
螺栓 BOLT	6.8	M16X40		94	13.6	
		M16X50		92	14.7	
		M16X60		20	3.5	
		M20X45		46	12.4	
		M20X55		102	30.1	
脚钉 STEP-BOLT	6.8	M16X180		15	4.9	
		M20X200		2	1.2	
垫圈 WASHER	Q235	-3(ø17.5)	规格X个数 SPECS X Qty	8	0.1	
		-4(ø17.5)		16	0.2	
		-4(ø22)		4	0.1	
合计 TOTAL (kg)			80.8 kg			
总计 SUM TOTAL (kg)			1191.2 kg			

构件明细表
BILL OF MATERIAL

编号 NO.	规格 SPECS	长度(mm) LENGTH	数量 Qty	重量 WEIGHT(kg) 一件 PIECE	重量 WEIGHT(kg) 合计 SUM	备注 REMARK
101	Q345L110X8	7875	1	106.57	106.6	带脚钉 WITH STEP BOLTS
102	Q345L110X8	7875	1	106.57	106.6	
103	Q345L110X8	7875	1	106.57	106.6	
104	Q345L110X8	7875	1	106.57	106.6	
105	Q345L80X6	430	4	3.17	12.7	铲背 HEEL FILLETING
106	Q345-6X90	430	4	1.82	7.3	
107	Q345-6X90	430	4	1.82	7.3	
108	L75X5	2517	2	14.64	29.3	切角 CHAMFERED
109	L75X5	2517	2	14.64	29.3	
110	L63X5	1158	2	5.58	11.2	
111	L56X5	2211	2	9.40	18.8	切角 CHAMFERED
112	L56X5	2211	2	9.40	18.8	
113	L50X5	2097	2	7.91	15.8	切角 CHAMFERED
114	L50X5	2097	2	7.91	15.8	
115	L50X5	1795	2	6.77	13.5	切角 CHAMFERED
116	L50X5	1795	2	6.77	13.5	
117	L50X5	863	2	3.25	6.5	
118	-6X216	485	4	4.93	19.7	
119	-6X206	250	4	2.43	9.7	
120	L75X5	3120	2	18.15	36.3	切角 CHAMFERED
121	L75X5	3120	2	18.15	36.3	
122	L63X5	1899	2	9.16	18.3	
123	L56X5	2681	2	11.40	22.8	切角 CHAMFERED
124	L56X5	2681	2	11.40	22.8	
125	L50X5	2429	2	9.16	18.3	切角 CHAMFERED
126	L50X5	2429	2	9.16	18.3	
127	L50X5	2049	2	7.72	15.4	切角 CHAMFERED
128	L50X5	2049	2	7.72	15.4	
129	L50X5	1170	2	4.41	8.8	
130	-6X242	322	4	3.67	14.7	
131	-6X160	214	4	1.61	6.4	
132	-6X160	214	4	1.61	6.4	
133	-6X217	250	4	2.56	10.2	
134	L50X5	1524	2	5.75	11.5	
135	-6X160	359	4	2.71	10.8	卷边50mm CRIMPED 50mm
136	L50X5	2307	2	8.70	17.4	
137	-6X163	376	4	2.89	11.6	卷边50mm CRIMPED 50mm
138	Q345-20X340	340	4	18.15	72.6	电焊 WELDING
139	Q345-10X391	302	4	9.27	37.1	电焊 WELDING
140	Q345-10X135	315	4	3.34	13.4	电焊 WELDING
141	Q345-10X255	304	4	6.09	24.4	电焊 WELDING
142	-2X60	180	8	0.17	1.4	垫块 CUSHION BLOCK
143	-12X60	60	4	0.34	1.4	垫块 CUSHION BLOCK
144	-8X60	120	2	0.45	0.9	垫块 CUSHION BLOCK
145	-10X50	50	4	0.20	0.8	垫块 CUSHION BLOCK
146	-14X50	50	4	0.27	1.1	垫块 CUSHION BLOCK
合计 TOTAL (kg)		1110.4 kg				

说明:
1、材料如无特别说明，杆件采用Q235B。
2、脚钉采用六角螺头型。

Specification:
1. If not specified, the member materials use Q235B.
2. The type of step-bolt is hexagon.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 3 A

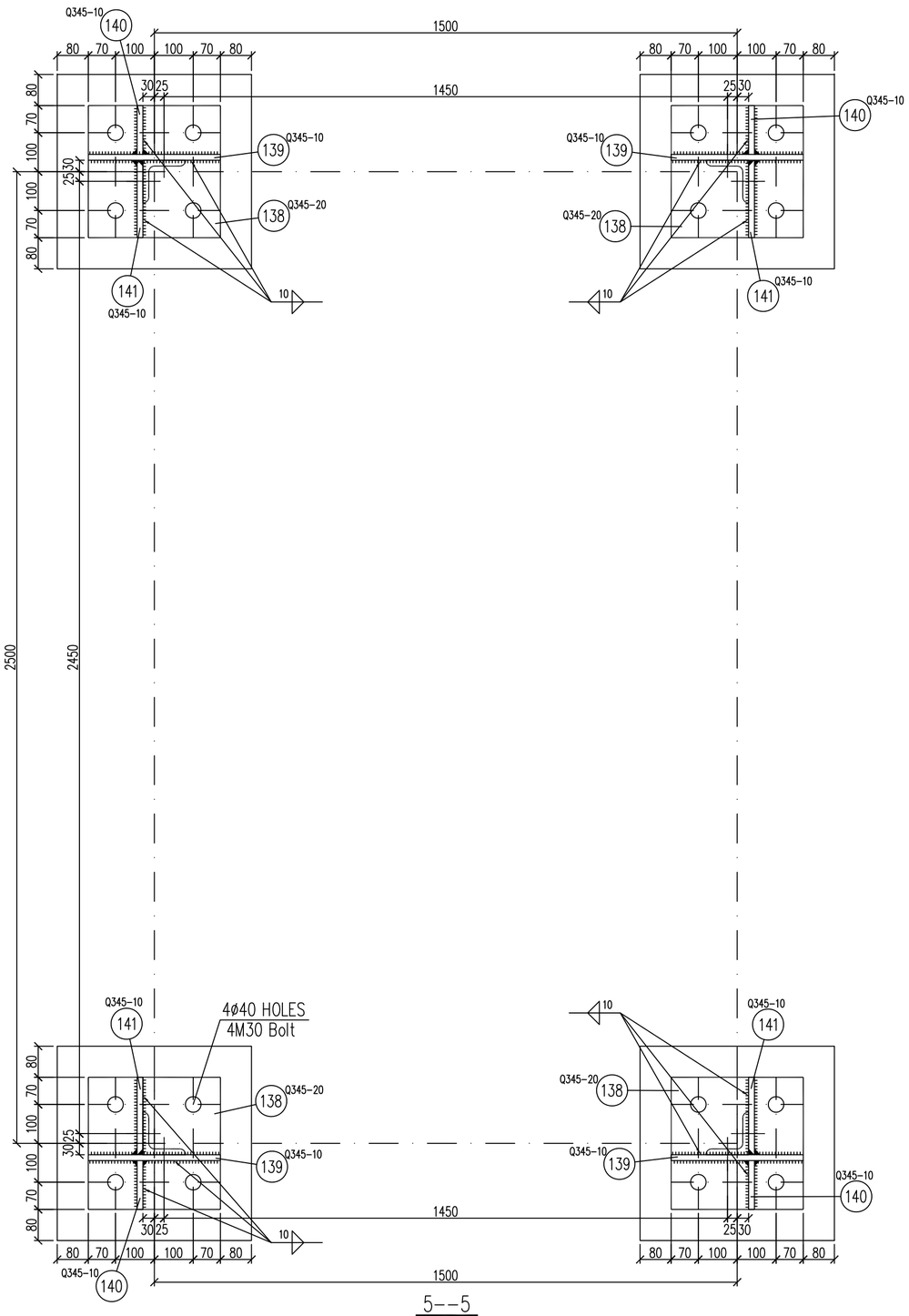
专业 Speciality	会签 Sign	日期 Date
变电一次		18.11.20

	姓名 Name	日期 Date
设计 Designed	龙金来	18.11.20
校核 Checked	石磊 古超城	18.11.20
审核 Reviewed	邹清清	18.11.20
批准 Approved	杨飞	18.11.20
比例 Scale	1:20	图号 DWG NO. 30-B003701S-T0306-003

直流接地极线构架腿身结构图(1/2) (1)
Extension Leg for frame of grounding electrode line(1/2) (1)

版本 REV. A

594X841



5--5
1:10

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 4 A

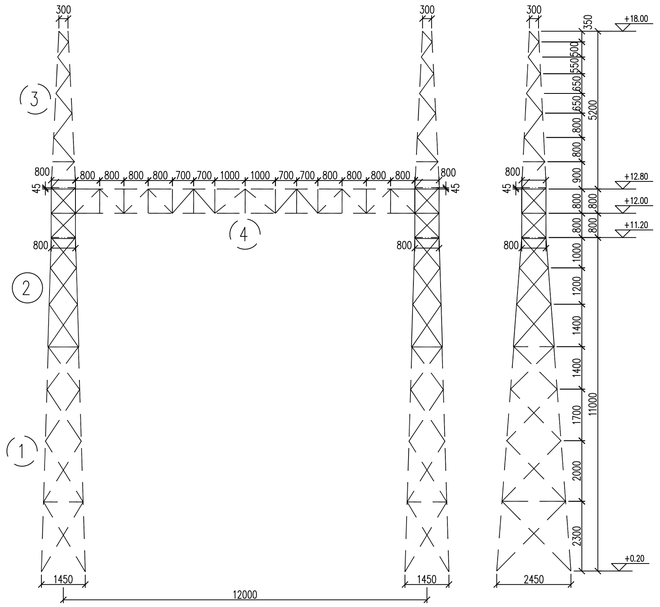
	姓名 Name	日期 Date
设计 Designed	龙金来	18.11.20
校核 Checked	毛惠平 古超城	18.11.20
审核 Reviewed	邹德清	18.11.20
批准 Approved	杨勇	18.11.20

直流接地极线构架腿身结构图(2/2) 1

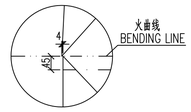
Extension Leg for frame of grounding electrode line(2/2) 1

比例 Scale	1:20	图号 DWG NO.	30-B003701S-T0306-004	版本 REV.	A

420X594



vertical view of frame of grounding electrode line



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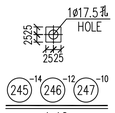
1:10



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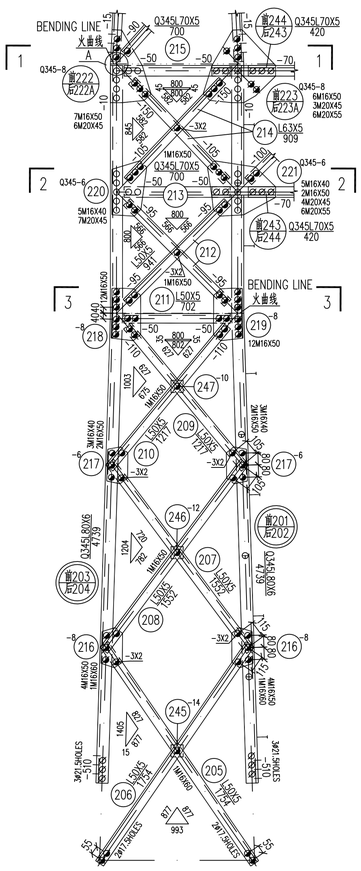
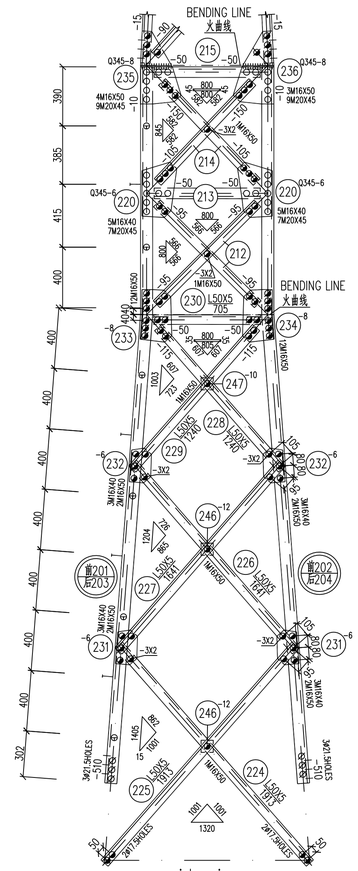


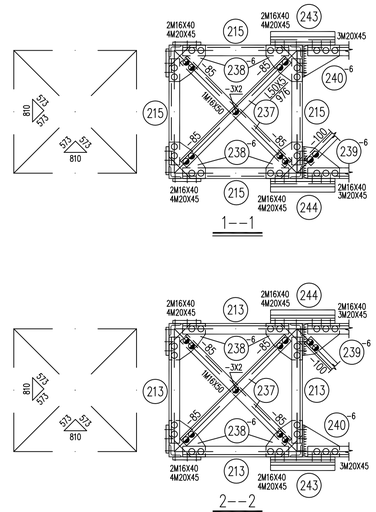
front view

正视图



side view

侧视图



1--1

2--2

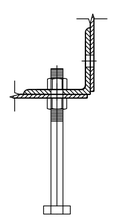
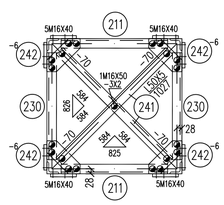


diagram of step-bolt in bolt hole



3--3

说明:

1、材料如无特别说明，杆件采用Q235B。

Specification:

1. If not specified, the member material uses Q235B.

螺栓、脚钉、垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备注 REMARK
螺栓 BOLT	6.8	M16X40		116	16.7	
		M16X50		213	34.1	
		M16X60		6	1.1	
		M20X45		136	36.7	
		M20X55		24	7.1	
脚钉 STEP-BOLT	6.8	M16X180		10	3.3	
		M20X200		1	0.6	
垫圈 WASHER	Q235	-3(ø17.5)	规格X个数 SPECS X Qty	54	0.6	
合计 TOTAL (kg)		100.2 kg				
总计 SUM TOTAL (kg)		730.1 kg				

构件明细表
BILL OF MATERIAL

编号 NO.	规格 SPECS	长度(mm) LENGTH	数量 Qty	重量 WEIGHT(kg) 一件 PIECE	重量 WEIGHT(kg) 合计 SUM	备注 REMARK
201	Q345L80X6	4739	1	34.95	35.0	切角 火曲 WITH STEP BOLTS HOT BEND
202	Q345L80X6	4739	1	34.95	35.0	火曲 HOT BEND
203	Q345L80X6	4739	1	34.95	35.0	火曲 HOT BEND
204	Q345L80X6	4739	1	34.95	35.0	火曲 HOT BEND
205	L50X5	1754	2	6.61	13.2	切角 CHAMFERED
206	L50X5	1754	2	6.61	13.2	
207	L50X5	1552	2	5.85	11.7	切角 CHAMFERED
208	L50X5	1552	2	5.85	11.7	
209	L50X5	1217	2	4.59	9.2	切角 CHAMFERED
210	L50X5	1217	2	4.59	9.2	
211	L50X5	702	2	2.65	5.3	
212	L50X5	941	8	3.55	28.4	
213	Q345L70X5	700	4	3.78	15.1	
214	L63X5	909	8	4.38	35.0	
215	Q345L70X5	700	4	3.78	15.1	
216	-8X140	265	4	2.33	9.3	
217	-6X137	244	4	1.57	6.3	
218	-8X194	346	2	4.22	8.4	火曲 HOT BEND
219	-8X194	346	2	4.22	8.4	火曲 HOT BEND
220	Q345-6X230	365	6	3.95	23.7	
221	Q345-6X365	449	2	7.72	15.4	
222	Q345-8X481	268	1	8.10	8.1	火曲 HOT BEND
222A	Q345-8X481	268	1	8.10	8.1	火曲 HOT BEND
223	Q345-8X481	478	1	14.44	14.4	火曲 HOT BEND
223A	Q345-8X481	478	1	14.44	14.4	火曲 HOT BEND
224	L50X5	1913	2	7.21	14.4	切角 CHAMFERED
225	L50X5	1913	2	7.21	14.4	
226	L50X5	1641	2	6.19	12.4	切角 CHAMFERED
227	L50X5	1641	2	6.19	12.4	
228	L50X5	1240	2	4.67	9.3	切角 CHAMFERED
229	L50X5	1240	2	4.67	9.3	
230	L50X5	705	2	2.66	5.3	
231	-6X139	230	4	1.51	6.0	
232	-6X139	230	4	1.51	6.0	
233	-8X209	346	2	4.54	9.1	火曲 HOT BEND
234	-8X209	346	2	4.54	9.1	火曲 HOT BEND
235	Q345-8X481	268	2	8.10	16.2	火曲 电焊 HOT BEND WELDING
236	Q345-8X481	268	2	8.10	16.2	火曲 电焊 HOT BEND WELDING
237	L50X5	976	4	3.68	14.7	
238	-6X134	339	8	2.14	17.1	
239	-6X210	235	2	2.32	4.6	电焊 WELDING
240	-6X210	235	2	2.32	4.6	电焊 WELDING
241	L50X5	1027	2	3.87	7.7	
242	-6X99	309	4	1.44	5.8	
243	Q345L70X5	420	2	2.27	4.5	
244	Q345L70X5	420	2	2.27	4.5	
245	-14X50	50	2	0.27	0.5	垫块 CUSHION BLOCK
246	-12X50	50	6	0.24	1.4	垫块 CUSHION BLOCK
247	-10X50	50	4	0.20	0.8	垫块 CUSHION BLOCK
合计 TOTAL (kg)		629.9 kg				

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	龙金来	18.11.20
校核 Checked	毛惠平 古韬城	18.11.20
审核 Reviewed	毛伟清	18.11.20
批准 Approved	杨少芳	18.11.20
比例 Scale	1:20	

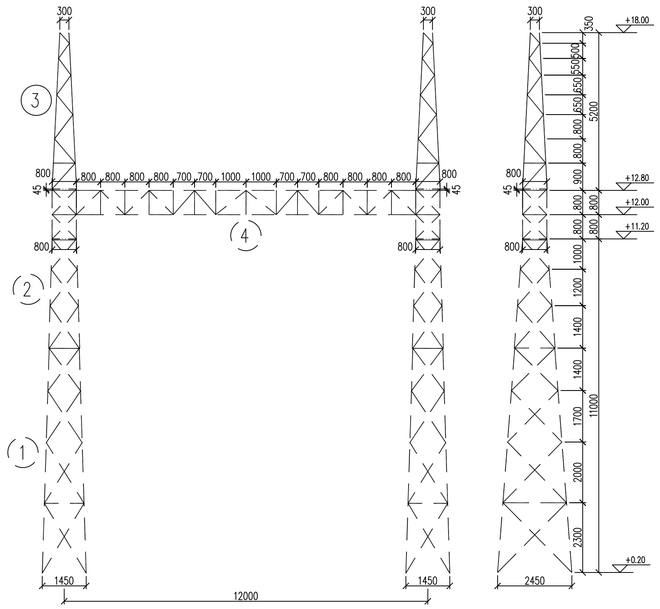
直流接地极线构架腿身结构图 (2)

Extension Leg for frame of grounding electrode line (2)

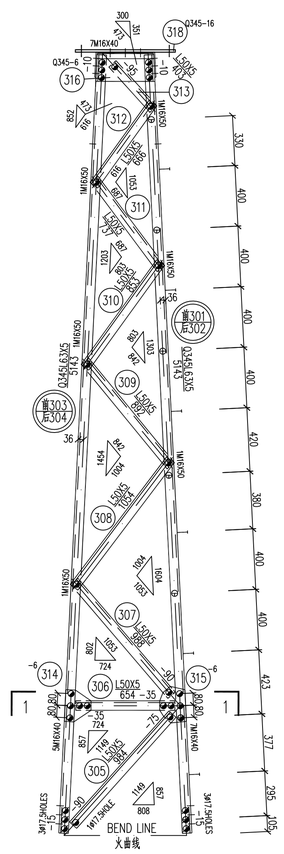
图号 DWG NO. 30-B003701S-T0306-005

版本 REV. A

594X841

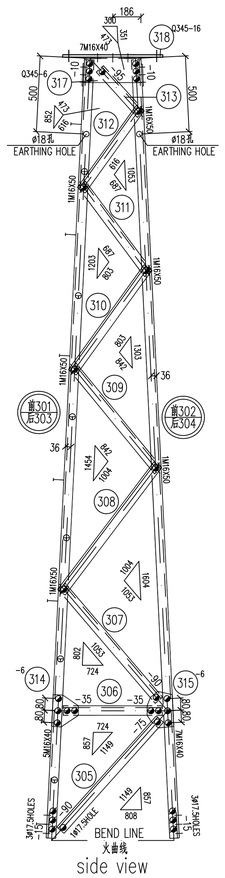


vertical view of frame of grounding electrode line



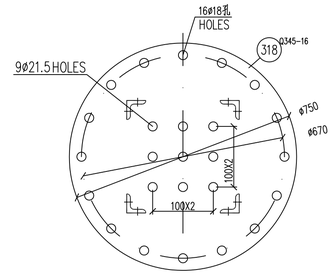
front view

正视图



side view

侧视图



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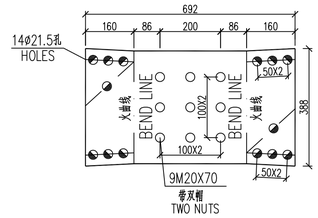

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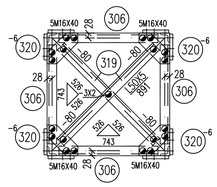

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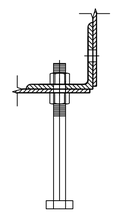


diagram of step-bolt in bolt hole

构件明细表
BILL OF MATERIAL

编号 NO.	规格 SPECS	长度(mm) LENGTH	数量 Qty	重量 WEIGHT(kg) 一件 PIECE	重量 WEIGHT(kg) 合计 SUM	备注 REMARK
301	Q345L63X5	5143	1	24.80	24.8	带脚钉 WITH STEP BOLTS
302	Q345L63X5	5143	1	24.80	24.8	
303	Q345L63X5	5143	1	24.80	24.8	
304	Q345L63X5	5143	1	24.80	24.8	
305	L50X5	984	4	3.71	14.8	
306	L50X5	654	4	2.47	9.9	
307	L50X5	988	4	3.72	14.9	切角 CHAMFERED
308	L50X5	1054	4	3.97	15.9	
309	L50X5	892	4	3.36	13.4	切角 CHAMFERED
310	L50X5	853	4	3.22	12.9	
311	L50X5	737	4	2.78	11.1	切角 CHAMFERED
312	L50X5	666	4	2.51	10.0	
313	L50X5	403	4	1.52	6.1	切角 CHAMFERED
314	-6X172	210	4	1.70	6.8	
315	-6X172	241	4	1.95	7.8	
316	Q345-6X180	394	2	3.34	6.7	
317	Q345-6X388	692	1	12.65	12.6	火曲 HOT BEND
318	Q345-16X750	750	1	70.65	70.7	
319	L50X5	891	2	3.36	6.7	
320	-6X99	287	4	1.34	5.4	
合计 TOTAL (kg)					324.9 kg	

螺栓、脚钉、垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备注 REMARK
螺栓 BOLT	6.8	M16X40		76	11.0	
		M16X50		24	3.8	
		M20X70		9	3.5	双螺母 TWO NUTS
脚钉 STEP-BOLT	6.8	M16X180		12	3.9	
垫圈 WASHER	Q235	-3(ø17.5)	规格X个数 SPECS X Qty	2	0.1	
合计 TOTAL (kg)				22.3 kg		
总计 SUM TOTAL (kg)				347.2 kg		

说明：

1、材料如无特别说明，杆件采用Q235B。

Specification:

1. If not specified, the member material uses Q235B.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	庞金来	18.11.20
校核 Checked	毛电子 古超帆	18.11.20
审核 Reviewed	邹清福	18.11.20
批准 Approved	杨浮	18.11.20
比例 Scale	1:20	

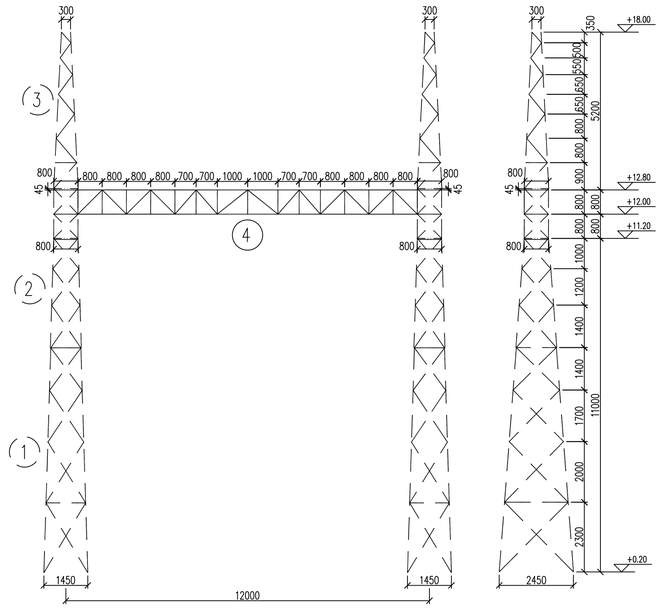
直流接地极构架身部结构图 (3)

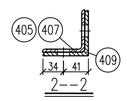
Superstructure for frame of grounding electrode line (3)

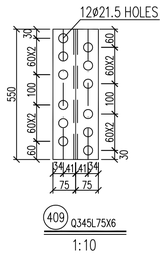
图号 DWG NO. 30-B003701S-T0306-006 版本 REV. A

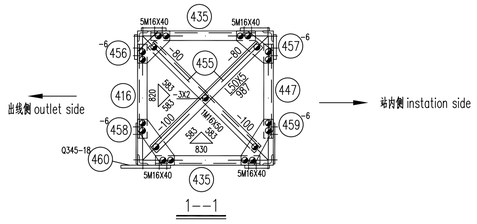
专业 Speciality	会签 Sign	日期 Date
变电一次	郑煜	18.11.20

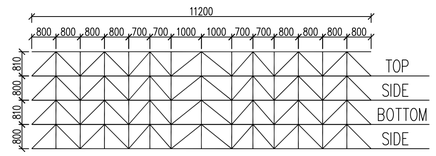
594X841

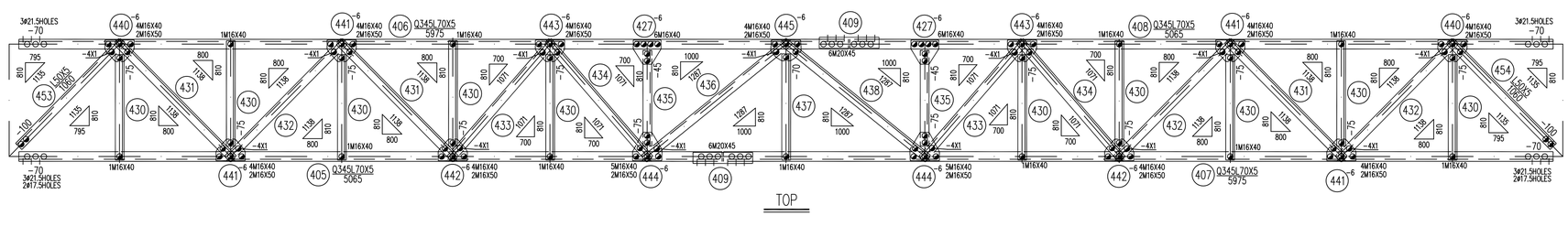


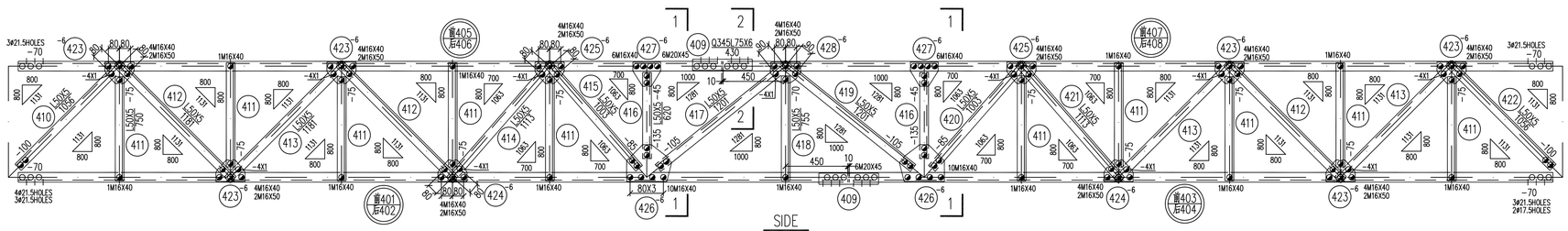


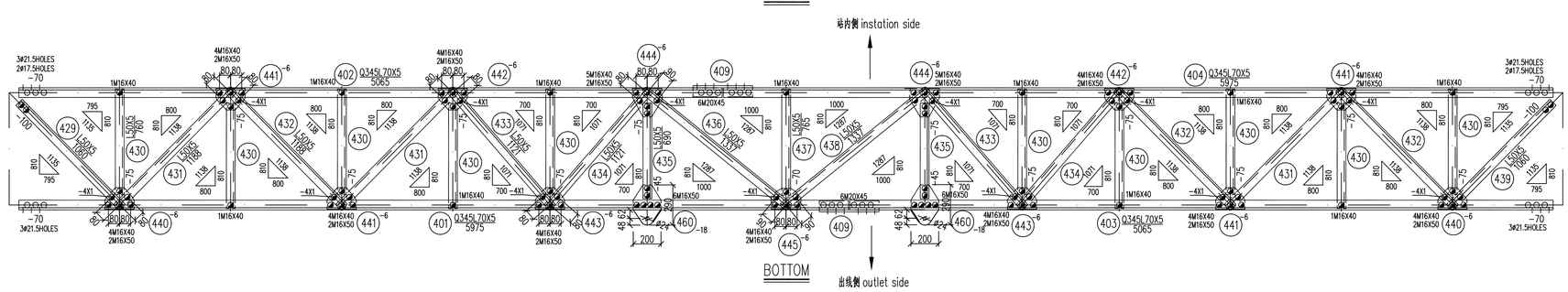


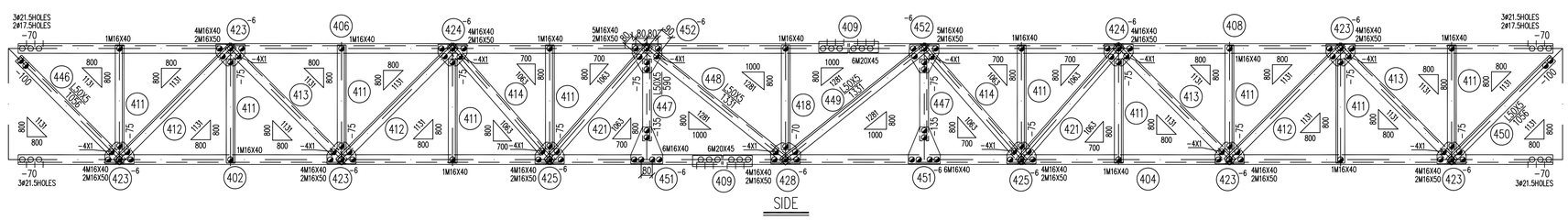












构 件 明 细 表
BILL OF MATERIAL

编号 NO.	规 格 SPECS	长度(mm) LENGTH	数量 Qty	WEIGHT(kg) 重量 一件 PIECE	合 计 SUM	备 注 REMARK
457	-6X108	323	2	1.64	3.3	
458	-6X114	388	2	2.08	4.2	
459	-6X122	370	2	2.13	4.3	
460	Q345-18X290	200	2	8.20	16.4	
合 计 TOTAL (kg)			786 kg			

螺栓、脚钉、垫圈明细表
BILL OF BOLT.STEP-BOLT.WASHER

名称 NAME	级别 GRADE	规 格 SPECS	符号 SAMPLE	数量 Qty	重量 WEIGHT(kg)	备 注 REMARK
螺 栓 BOLT	6.8	M16X40		346	49.9	
		M16X50		114	18.2	
		M20X45		48	13.0	
脚钉 STEP-BOLT						
垫 圈 WASHER	Q235	-3(Ø17.5)	规格X个数 SPECS X Qty	4	0.1	
		-4(Ø17.5)		50	0.7	
合 计 TOTAL (kg)			81.9kg			
总 计 SUM TOTAL (kg)			867.9 kg			

构 件 明 细 表
BILL OF MATERIAL

编号 NO.	规 格 SPECS	长度(mm) LENGTH	数量 Qty	WEIGHT(kg) 重量 一件 PIECE	合 计 SUM	备 注 REMARK
401	Q345L70X5	5975	1	32.25	32.2	
402	Q345L70X5	5065	1	27.34	27.3	
403	Q345L70X5	5065	1	27.34	27.3	
404	Q345L70X5	5975	1	32.25	32.2	
405	Q345L70X5	5065	1	27.34	27.3	
406	Q345L70X5	5975	1	32.25	32.2	
407	Q345L70X5	5975	1	32.25	32.2	
408	Q345L70X5	5065	1	27.34	27.3	
409	Q345L75X6	430	4	2.97	11.9	铲板 ARC SHOVELED
410	L50X5	1056	1	3.98	4.0	切角 CHAMFERED
411	L50X5	750	20	2.83	56.6	
412	L50X5	1181	6	4.45	26.7	
413	L50X5	1181	6	4.45	26.7	切角 CHAMFERED
414	L50X5	1113	3	4.20	12.6	切角 CHAMFERED
415	L50X5	1003	1	3.78	3.8	
416	L50X5	620	2	2.34	4.7	
417	L50X5	1201	1	4.53	4.5	切角 CHAMFERED
418	L50X5	755	2	2.85	5.7	
419	L50X5	1201	1	4.53	4.5	
420	L50X5	1003	1	3.78	3.8	切角 CHAMFERED
421	L50X5	1113	3	4.20	12.6	
422	L50X5	1056	1	3.98	4.0	
423	-6X155	210	12	1.53	18.4	
424	-6X155	210	4	1.53	6.1	
425	-6X155	210	4	1.53	6.1	
426	-6X265	323	2	4.03	8.1	电焊 WELDING
427	-18X185	200	4	1.65	6.6	
428	-6X150	221	2	1.56	3.1	
429	L50X5	1060	1	4.00	4.0	切角 CHAMFERED
430	L50X5	760	20	2.87	57.4	
431	L50X5	1188	6	4.48	26.9	
432	L50X5	1188	6	4.48	26.9	切角 CHAMFERED
433	L50X5	1121	4	4.23	16.9	切角 CHAMFERED
434	L50X5	1121	4	4.23	16.9	
435	L50X5	690	4	2.60	10.4	
436	L50X5	1337	2	5.04	10.1	切角 CHAMFERED
437	L50X5	765	2	2.88	5.8	
438	L50X5	1337	2	5.04	10.1	
439	L50X5	1060	1	4.00	4.0	
440	-6X155	210	4	1.53	6.1	
441	-6X155	210	8	1.53	12.2	
442	-6X155	210	4	1.53	6.1	
443	-6X155	210	4	1.53	6.1	
444	-6X205	215	4	2.08	8.3	
445	-6X150	216	2	1.53	3.1	
446	L50X5	1056	1	3.98	4.0	切角 CHAMFERED
447	L50X5	590	2	2.22	4.4	
448	L50X5	1331	1	5.02	5.0	切角 CHAMFERED
449	L50X5	1331	1	5.02	5.0	
450	L50X5	1056	1	3.98	4.0	
451	-6X230	265	2	2.87	5.7	电焊 WELDING
452	-6X205	215	2	2.08	4.2	
453	L50X5	1060	1	4.00	4.0	切角 CHAMFERED
454	L50X5	1060	1	4.00	4.0	
455	L50X5	987	4	3.72	14.9	
456	-6X99	301	2	1.40	2.8	

说明：

1、材料如无特别说明，杆件采用Q235B。

2、除挂线点隔面布置交叉材，其余隔面无交叉材。

3、横梁需设0.3%的预拱度。

Specification :

1. If not specified, the member material uses Q235B.
2. No cross members are placed in transverse plane, except the hanging point.
3. The beam shall have a pre-arch of 0.3%.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 0 6 - 0 0 7 A

	姓名 Name	日期 Date
设计 Designed	龙金林	18.11.20
校核 Checked	毛惠千 古超卿	18.11.20
审核 Reviewed	邹清福	18.11.20
批准 Approved	杨岩	18.11.20

直流接地极构架梁结构图 (4)

Beam structure for frame of grounding electrode line (4)

比例 Scale: 1:20 | 图号 DWG NO.: 30-B003701S-T0306-007 | 版本 REV.: A

594X841